Introduction to NLP

AIAA 4051 Lecture 9

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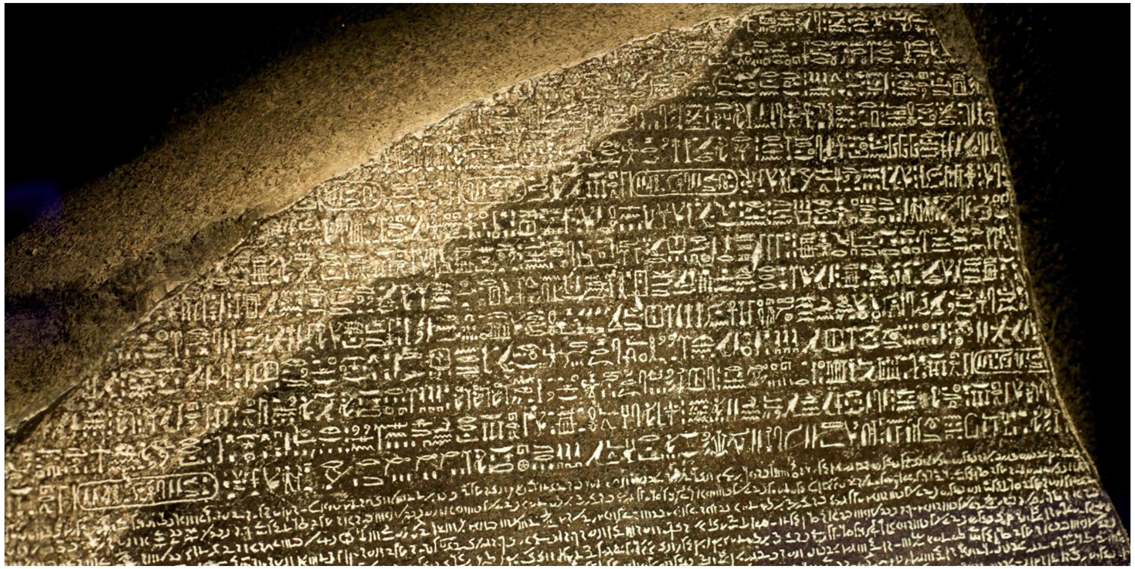
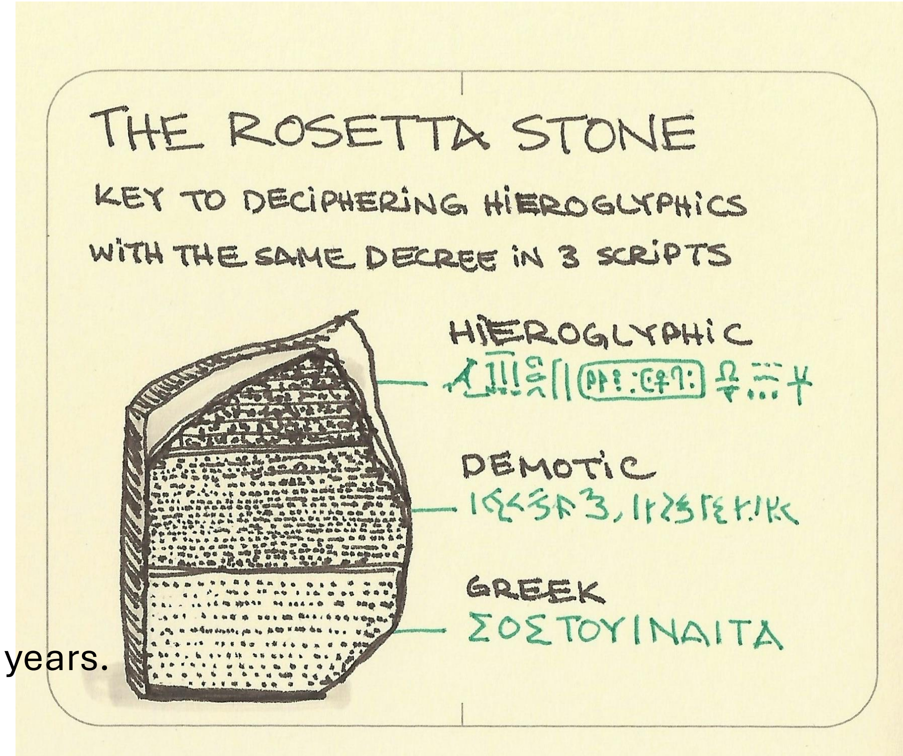
Machine translation in ancient Greece

- There is a Rosetta Stone 2000 years old, discovered in 1799 in Egypt.





Royal orders in three different languages:
Hieroglyphic: Used for formal priestly decrees, unknown for 1400 years.
Demotic: The common "native" script of daily Egyptian life.
Greek: something modern human can understand.

How to read the dead language using a known language?

Image source:
https://www.history.com/this-day-in-history/july-15/rosetta-stone-foun
https://sketchplanations.com/the-rosetta-stone

Machine translation in ancient Greece

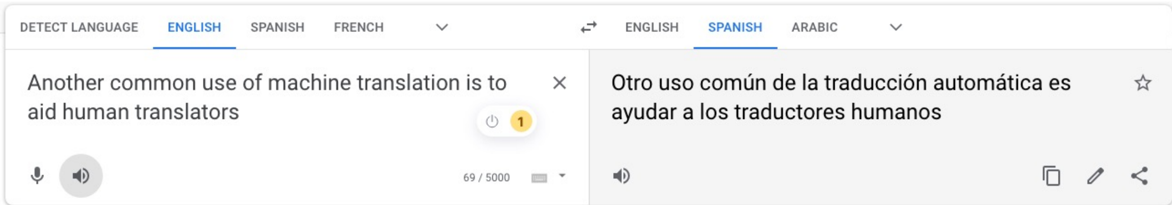
- Tricks to translation
  - check patterns in the unknown language and make statistics.
  - words with similar statistics and in similar locations are likely to be a pair of translation.
  - need to try several alignments and find the most consistent ones.
    - A greek word should not be translated into many different unknown patterns.

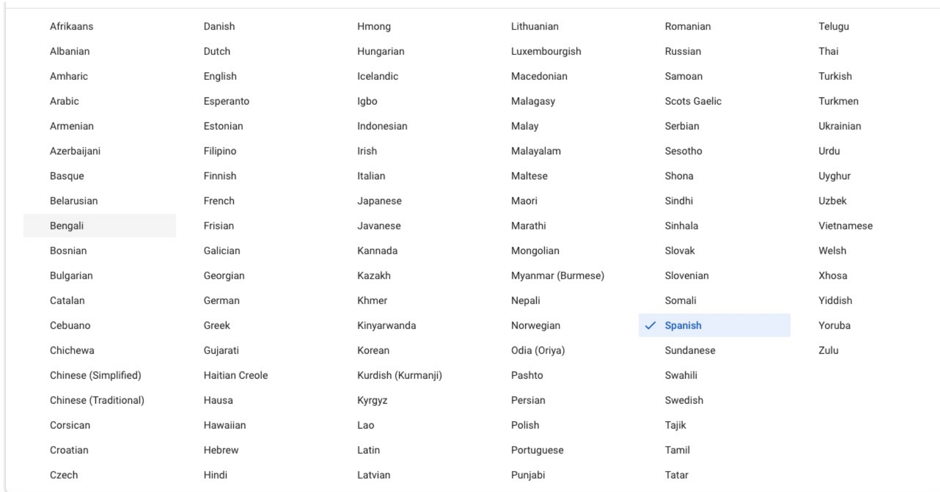
- A modern Rosetta stone
  - **Hansards**—the official transcripts of the Canadian Parliament, which are legally required to be published in both **English** and **French**.
  - For a computer scientist, the Canadian Hansards *are* the modern Rosetta Stone: a massive, perfectly aligned dataset that allows a machine to learn how to translate by simply "counting" which words tend to appear together.


REVISED EDITION

Neural networks for language modeling

- Translating sentences in one language to those in another.





- Was an active research area in NLP.
  - No longer so due to LLM.
- Used to have large-scale parallel training corpora and computing infrastructures.
  - but now LLM learns to generate a language given any context.

Machine translations before and now

Feature	IBM Model 1 (The 1990s)	Large Language Models (2026)
Fundamental Unit	**The Individual Word.** It treats a sentence like a bucket of marbles.	**The Context/Token.** It understands how words relate across entire books.
Logic Type	**Statistical Frequency.** "How often does *A* appear with *B*?"	**Semantic Reasoning.** "What does the speaker *intend* to say here?"
The "Idiom" Test	**Fails.** "Kick the bucket" becomes "Coup de pied au seau" (literal kicking of a pail).	**Succeeds.** Translates "Kick the bucket" to "Casser sa pipe" (French equivalent for dying).
Cultural Awareness	**None.** It doesn't know what a "bucket" or a "country" is.	**High.** Knows to change "miles" to "kilometers" or "dollars" to "euros" for a local audience.
Constraint Handling	**Impossible.** You cannot tell Model 1 to "be shorter" or "use a formal tone."	**Native.** Can be prompted: "Translate this email into Japanese but use *Keigo* (honorifics)."

Let’s look at the languages

- The challenges of machine translation researchers faced before.
- Many differences between two languages on many levels.
  - lexical level
    - “*bass*” in English can be translated into “*bajo*” /ˈba:ho/ (a musical instrument) or “*lubina*” /luˈbina:/ (a fish) in Spanish.
    - “wall” in English can be translated into “*Wand*” (wall inside a building), or “*Mauer*” /mauwa/ (wall outside a building) in German.
    - ”brother” in English can be translated into “哥哥” /gege/ (older brother) or “弟弟” /didi/ (younger brother) in Chinese.
  - syntactic level
    - In English: no gender for adjectives; in French/Spanish, adjectives can have genders.

Let’s look at the languages

- Many differences between two languages on many levels.
  - syntactic level (word ordering)
    - SVO (Subject-verb-object): English
      - *“He adores listening to music”*
    - SOV (Subject-objective-verb): Japanese
      - word-to-word translation from Japanese

        “he music to listening adores”
    - different places for pre-position phrases (PP)

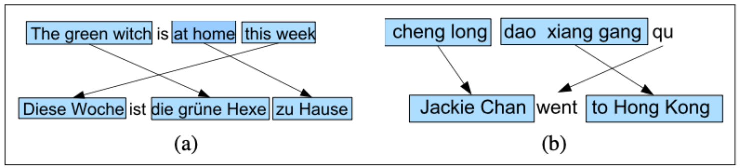


Figure 11.1 Examples of other word order differences: (a) In German, adverbs occur in initial position that in English are more natural later, and tensed verbs occur in second position. (b) In Mandarin, preposition phrases expressing goals often occur pre-verbally, unlike in English.

English: *He wrote a letter to a friend*

Japanese: *tomodachi ni tegami-o kaita*

friend to letter wrote

Arabic: *katabt risāla li ṡadq*

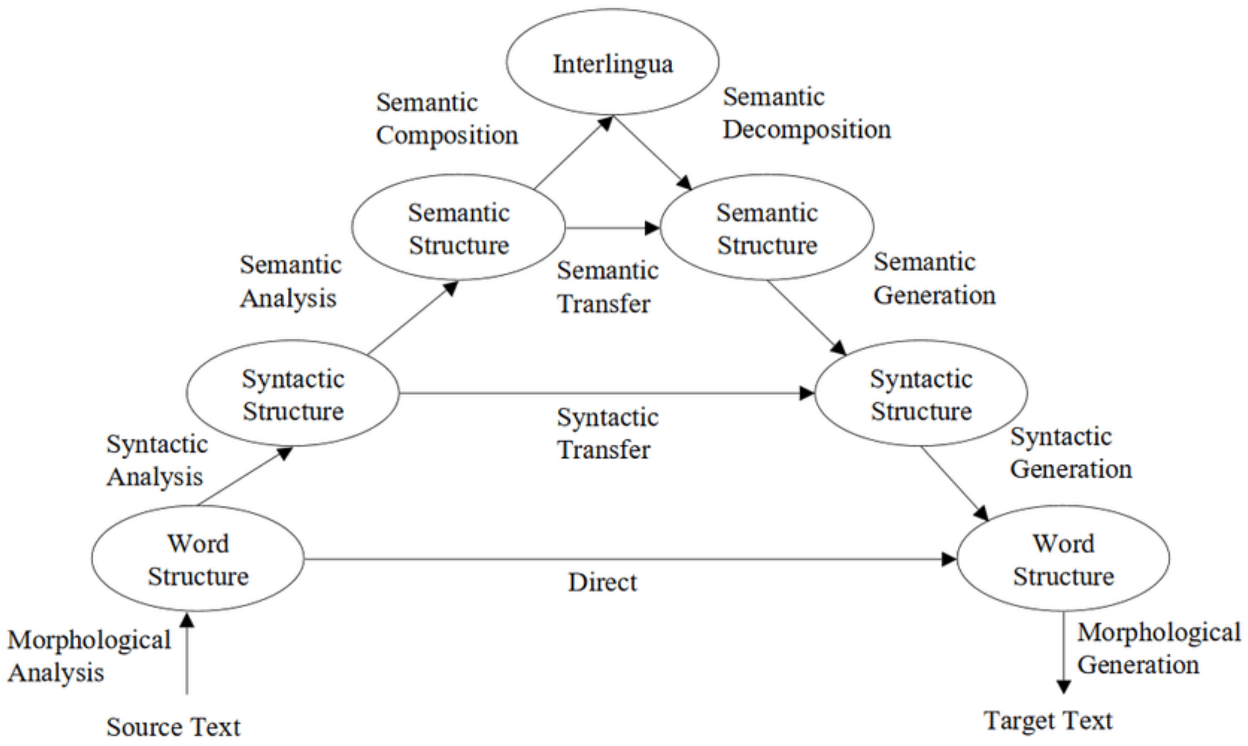
wrote letter to friend

- different adjective-noun ordering

Spanish *bruja verde* **English** *green witch*

Several levels of translations

- Vouquois* /’va:kua:/ triangle



* it is a French word

Rule-based translation

Direct translation

English:	Mary didn't slap the green witch
Spanish:	*Maria no dió una bofetada a la bruja verde*
Word-by-word translation:	Mary not gave a slap to the witch green

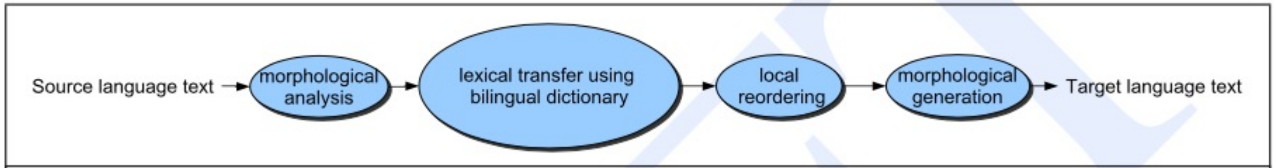


Figure 25.5 Direct machine translation. The major component, indicated by size here, is the bilingual dictionary.

Local ordering is for a phrase only

Input:	Mary didn't slap the green witch
After 1: Morphology	Mary DO-PAST not slap the green witch
After 2: Lexical Transfer	Maria PAST no dar una bofetada a la verde bruja
After 3: Local reordering	Maria no dar PAST una bofetada a la bruja verde
After 4: Morphology	Maria no dió una bofetada a la bruja verde

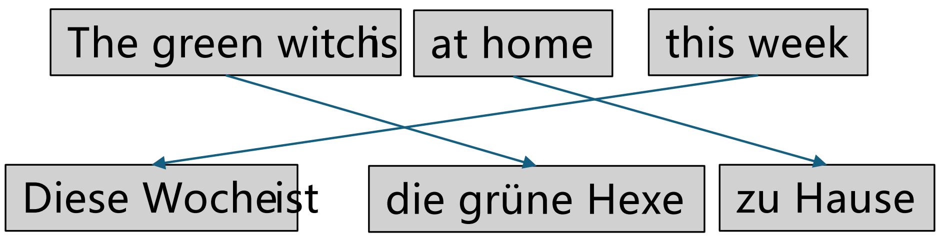
Figure 25.6 An example of processing in a direct system

Image source: statistical language processing, 2nd Edition

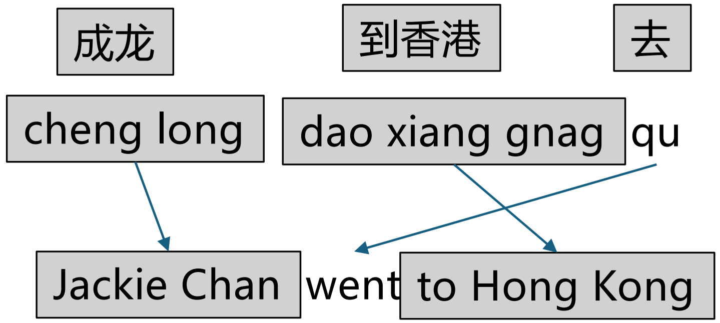
Rule-based translation

- Direct method
  - drawbacks: can't handle different orders of larger linguistic units, such as phrases.
  - Where to place a phrase (such as prepositions) in a different language needs more global information of the sentence.

English to German



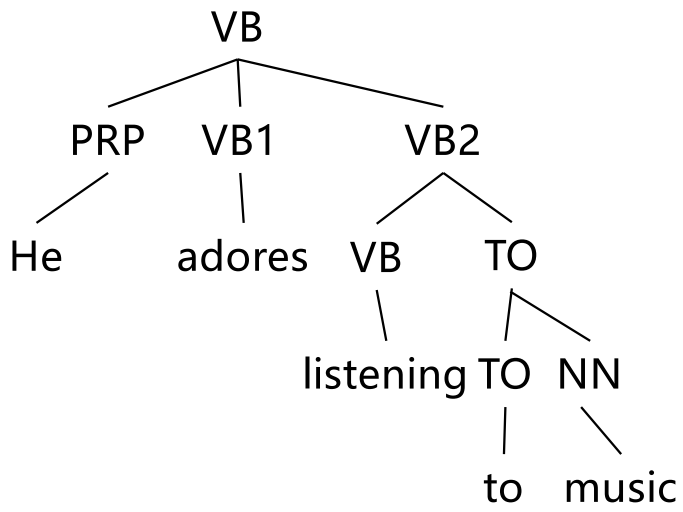
Chinese to English



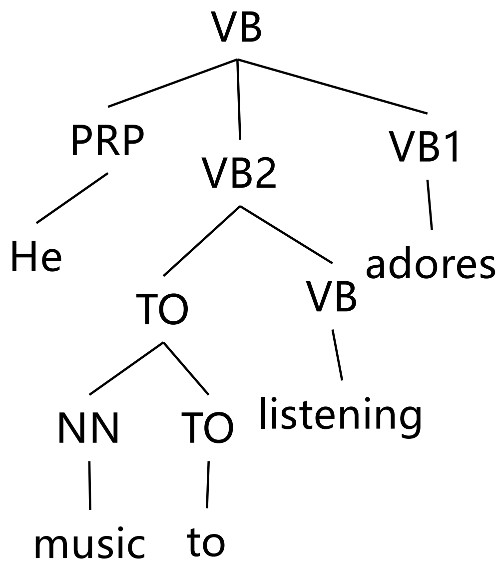
Rule-based translation

- Transfer method
    - syntactic transfer: transforming a parsing tree into another.
    - lexical transfer: translate words to words.

English parse tree



Japanese parse tree



Metrics of translation

- Metrics
  - Fluency and faithfulness.
  - Hebrew: *adonai roi* to English "*the Lord is my shepherd*"
  - One translation: "*the Lord will look after me*" (fluent but not too faithful)
  - Another translation "*the Lord is for me like somebody who looks after animals with cotton-like hair*" (faithful but not fluent)

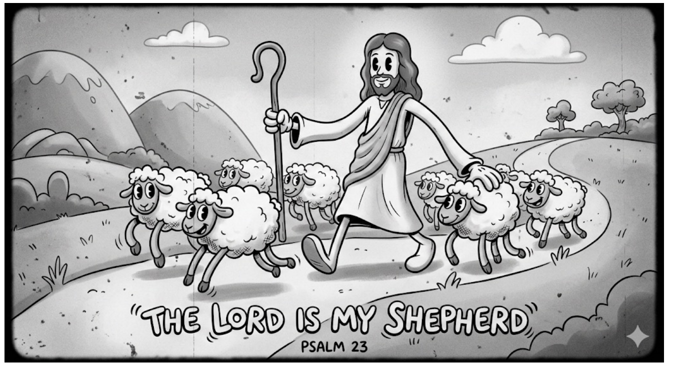


Image source: Gemini

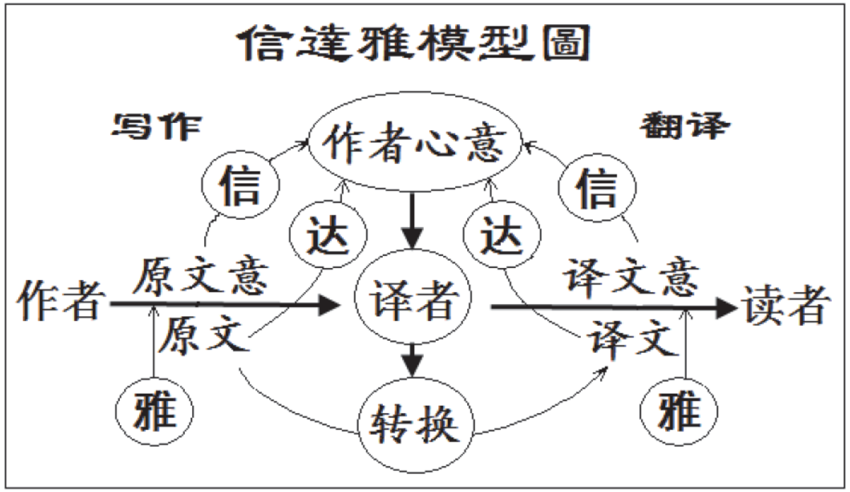
Metrics of translation

- Need some trade-off between the two goals:
  - S: the source sentence; T: the translation

$$\text{best-translation } \hat{T} = \text{argmax}_T \text{ fluency(T) faithfulness(T,S)}$$

- More formally,

$$\text{best-translation } \hat{T} = \text{argmax}_T \text{ P(T) } P(S|T)$$



“信达雅” are three standards of translation proposed by “严复”

•“信”（**faithfulness**）指意义不悖原文，即是译文要准确，不偏离，不遗漏，也不要随意增减意思；
•“达”（**expressiveness**）指不拘泥于原文形式，译文通顺明白；
•“雅”（**elegance**）则指译文时选用的词语要得体，追求文章本身的古雅，简明优雅。

Goal of translation

- Goals of MT
  - Fluency and faithfulness

$$\text{best-translation } \hat{T} = \text{argmax}_T \text{ P(T) } P(S|T)$$

  - More formally,
    - $T=E=(e_1, \ldots, e_I)$: English (the target language).
    - $S=F=(f_1, \ldots, f_J)$: a foreign language (Spanish/French/…).
    - Language model regarding the **fluency** of the translation $P(E)$
    - Translation model regarding the **faithfulness** of the translation $P(F|E)$
    - Using the Bayes theorem, *decoding* is defined as

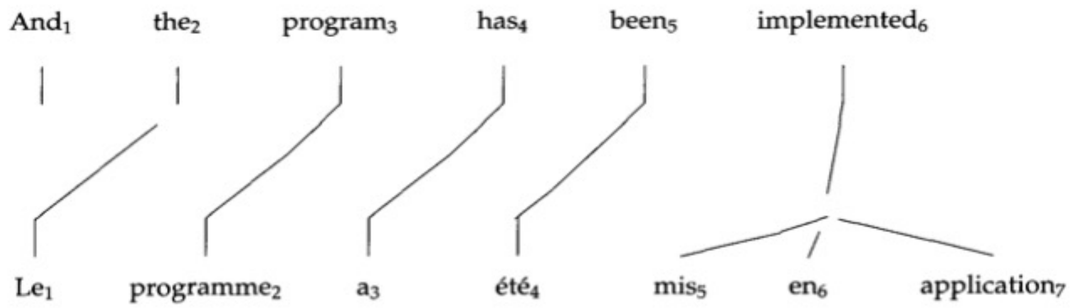
$$E^* = \text{argmax}_E P(E|F) = \text{argmax}_E \frac{P(F|E)P(E)}{P(F)} = \text{argmax}_E P(F|E)P(E)$$

discriminative model

generative model

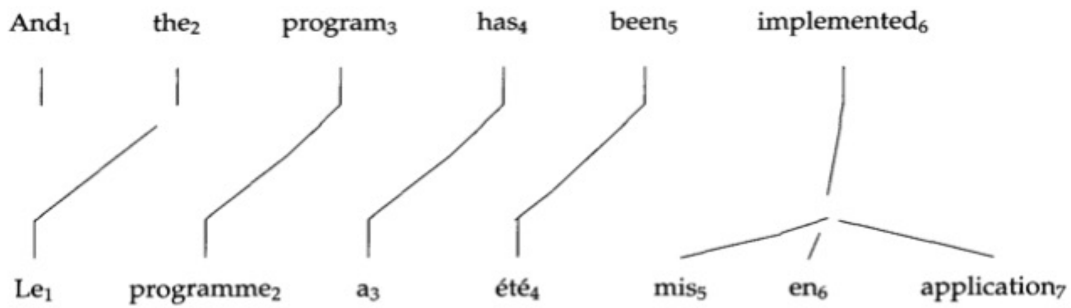
Alignments

- Word alignment: mapping words in *E* to words in *F*.
  - Multiple source (French) words can be mapped to one target (English) word.
  - Example:



  - Recording the mapping *A* directly: $a_1=2$, $a_2=3$, …, $a_7=6$.

Alignments



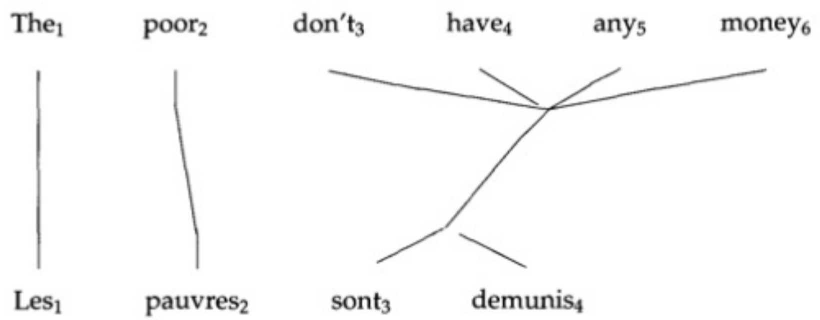
Alignment matrix

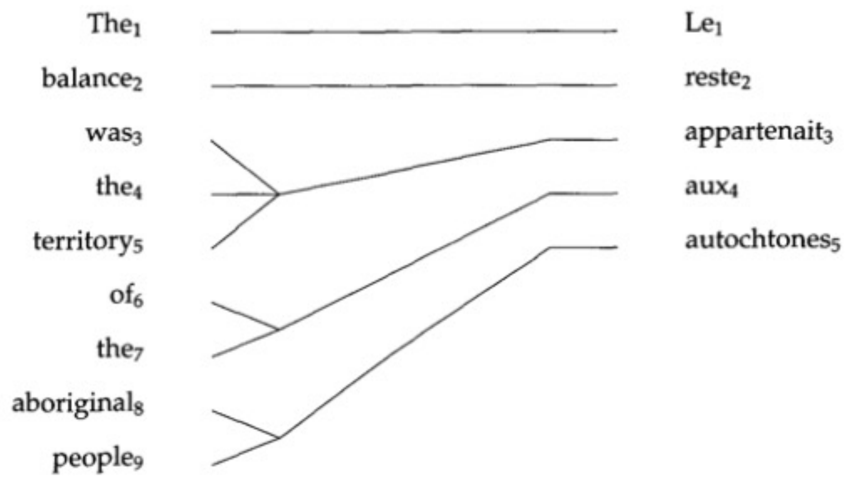
Source (French) / Target (English)	Le (f1)	programme (f2)	a (f3)	été (f4)	mis (f5)	en (f6)	application (f7)
the (e2)	X						
program (e3)		X					
has (e4)			X				
been (e5)				X			
implemented (e6)					X	X	X

Alignments

- One source word mapped to multiple target words.

- Many-to-many mapping

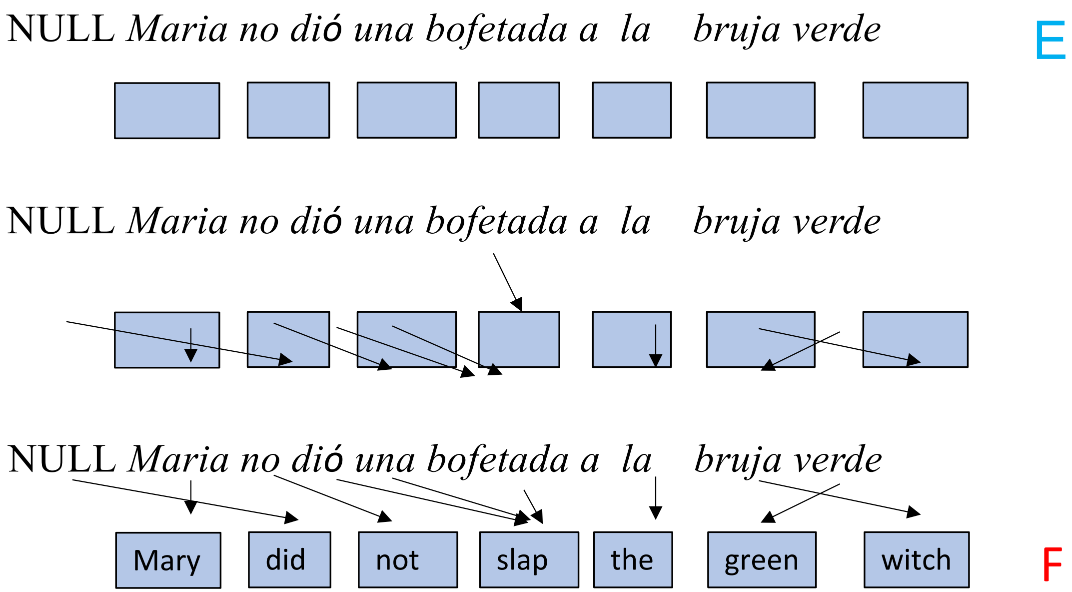




- Allow spurious word NULL for words that can't be mapped.

IBM Model 1 translator

- Published in *Brown, etc.1993* [1].
- Input: F; Output: translations E.
- Translation model $P(F|E) = \sum_{A} P(F,A|E)$
- Generative story
  1. generate length of the foreign language *J*;
  2. generate an alignment *A*;
  3. generate words $F=(f_1, \ldots, f_J)$ in the foreign language.



[1] The Mathematics of Statistical Machine Translation. Brown, etc. 1993 Computational Linguistics

IBM Model 1 translator

- Probability of generating a length J: a small positive number ϵ
- Probability of an alignment between I and J words: $P(A(I,J)) = \frac{1}{(I+1)^J}$
- Probability of the foreign word f_j given the aligned target word e_{a_j}

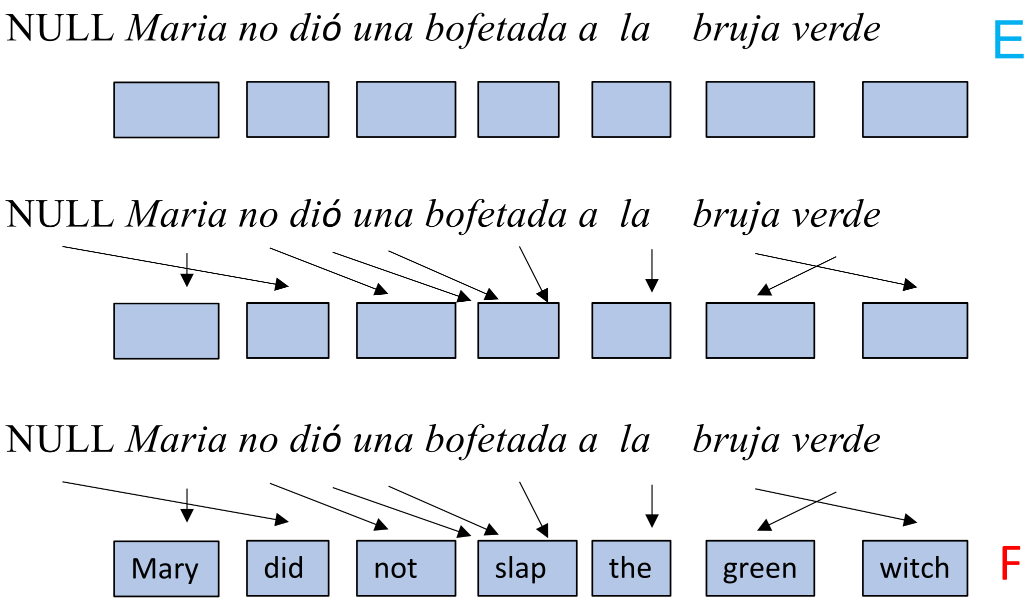
$$P(f_j|e_{a_j})$$

- Probability of the foreign sentence and an alignment A:

$$P(F,A|E) = \frac{\epsilon}{(I+1)^J}\prod_{j=1}^{J} P(f_j|e_{a_j})$$

- Probability of F given E:

$$P(F|E) = \sum_A P(F,A|E) = \sum_A \frac{\epsilon}{(I+1)^J}\prod_{j=1}^{J} P(f_j|e_{a_j})$$



Problems with Model 1

- **The Bag-of-Words Assumption (No Distortion):** Every possible alignment (mapping word i to word j) is equally likely, regardless of distance.
    - **The Reality:** In real language, word order matters. If you translate "The cat is small" to French "Le chat est petit," the words usually stay in a similar relative order. Model 1 ignores this entirely.
- **Independent Word Translation:** The choice of translating e_i to f_j depends only on that specific word pair and nothing else in the sentence.
    - **The Reality:** the word "bank" has the same probability of being "banque" (financial) or "rive" (river), regardless of whether the surrounding words are "money" or "water."
- **One-to-Many Alignment:** Each source word f_j must align to exactly **one** target word e_i. However, a single target word e_i can be the parent of multiple source words (fertility).
    - **The Reality:** Languages often have many-to-many relationships (e.g., idioms where three words in one language equal two in another).

Extra readings

- For alignment and machine translation, see Chap 13 of FSNLP.

- Jianhui Pang, Fanghua Ye, Derek Fai Wong, Dian Yu, Shuming Shi, Zhaopeng Tu, and Longyue Wang. 2025. Salute the Classic: Revisiting Challenges of Machine Translation in the Age of Large Language Models. Transactions of the Association for Computational Linguistics, 13:73–95.

- Zhixiao Zhao, Guangyao Sun, Chang Liu, Dongbo Wang. Research on machine translation of ancient books in the era of large language model. NPJ Heritage Science volume 13, Article number: 122 (2025)

Demon

Conclusion

- Machine translation is to turn texts in one language to another.
- There are rule-based and statistics-based approaches.
- LLM is changing how machine translation is done.

Quiz

1. Pen and paper; no compute or cellphone allowed.
2. Turn in your answer sheet when you leave the classroom.

- Q1 (T/F): An alignment matrix can represent all of one-to-one, one-to-many, and many-to-one mappings.
- Q2 (T/F): in statistics-based translation, P(E|F) can be used to describe the fluency of the translation E from the foreign language F.
- Q3 (T/F): the direct translation method can handle long-range dependencies when translating a short phrase.
- A: T, F, F